

2) $X_i \stackrel{iid}{\sim} \text{Gamma}(k, \lambda)$

$i = 1, \dots, n$

p.d.f of X_i is given by

$$f(x; k, \lambda) = \begin{cases} \frac{1}{\Gamma(k)} x^{k-1} \lambda^k \exp(-\lambda x), & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

Its moment generating function (M_X) is given by

$$M_{X_i}(t) = (1 - \lambda t)^{-k}$$

$$t < \frac{1}{\lambda}$$

~~$M_{X_i}(t)$~~

$$\text{Let } Y = \sum_{i=1}^n X_i$$

then

$$M_Y(t) = \prod_{i=1}^n (1 - \lambda t)^{-k}$$

$$= (1 - \lambda t)^{-nk}$$

So,

$$Y \sim \text{Gamma}(nk, \lambda)$$

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$$\sum_{i=1}^n X_i \sim \text{Gamma}(nk, \lambda)$$

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$$E\left[\frac{\sum_{i=1}^n X_i}{n}\right] = \frac{nk\lambda}{n} = k\lambda$$

$$\begin{aligned}
 V\left(\frac{\sum_{i=1}^n X_i}{n}\right) &= E\left[\left(\frac{Y}{n} - E\bar{X}_n\right)\left(\frac{Y}{n} - E\bar{X}_n\right)\right] \\
 &= E\left[\left(\frac{Y}{n} - \frac{EY}{n}\right)\left(\frac{Y}{n} - \frac{EY}{n}\right)\right] \\
 &= \frac{V(Y)}{n^2} = \frac{\alpha \lambda^2}{n^2}
 \end{aligned}$$

$$AB = \alpha \lambda^2$$

$$AB^2 = \frac{\alpha \lambda^2}{n^2}$$

$$B = \frac{\lambda}{n^2}$$

$$A = n^2 \alpha$$

So,

~~$$\bar{X}_n \sim \text{Gamma}(n^2 \alpha, \frac{\lambda}{n^2})$$~~

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$y_1, \dots, y_r \mid X \text{ ist Poisson}(X)$
 $X \sim \text{Gamma}(a, b)$

$$f(X \mid y_1, \dots, y_r)$$

$$\propto \pi(x) f(y_1, \dots, y_r \mid x)$$

$$= \frac{\pi(x) f(y_1, \dots, y_r \mid x)}{m(x)}$$

$$= \left(\frac{1}{\Gamma(a)} x^{a-1} b^{-a} \exp\left(-\frac{x}{b}\right) \right) \left(e^{-x} \frac{x^{y_1}}{y_1!} \right)$$

x

$$= \frac{1}{y_1! \Gamma(a)} x^{a+y_1-2} b^{-a} \exp\left(-\frac{x}{b/1+b}\right)$$

1) $X_n : n \geq 1$ are independent variables such that
 $X_i \sim \text{Bernoulli}(\theta)$ $\theta \in (0, 1)$

M.G.F of X_i is given by

$$M_{X_i}(\theta) = 1 - \theta + \theta e^\theta$$

M.G.F of $Y_N = \sum_{n=1}^N X_n$ is given by

$$M_{Y_N}(\theta) = \prod_{n=1}^N (1 - \theta + \theta e^\theta) = (1 - \theta + \theta e^\theta)^N$$

It is the M.G.F of a Binomial (N, θ) distribution

So, $Y_N \sim \text{Binomial}(N, \theta)$

$$\text{Now, } \bar{Y}_N = \frac{Y_N}{N}$$

$$\text{So, } E \bar{Y}_N = \frac{E Y_N}{N} = \frac{N\theta}{N} = \theta$$

$$E \bar{Y}_N^2 = N^{-2} E Y_N^2 = \frac{N(N-1)\theta^2 + N\theta}{N^2} = \frac{(N-1)\theta^2 + \theta}{N}$$

$$\begin{aligned} \text{Var}(\bar{Y}_N) &= E \bar{Y}_N^2 - (E \bar{Y}_N)^2 \\ &= \frac{(N-1)\theta^2 + \theta}{N} - \theta^2 = \frac{\theta(1-\theta)}{N} \end{aligned}$$

So, $\bar{Y}_N$ is a Binomial distribution with mean θ & variance $\frac{\theta(1-\theta)}{N}$

support of x_i is $\{0, 1\}$

support of x_N is $\{0, 1, \dots, N\}$

support of $\bar{x}_N$ is $\{0, \frac{1}{N}, \frac{2}{N}, \dots, 1\}$

2) $X_n : n \geq 1$ is a sequence of IID Random Variables such that

$$X_i \stackrel{\text{IID}}{\sim} \text{Gamma}(\alpha, \lambda)$$

MGF of X_i is given by

$$M_{X_i}(t) = (1 - \lambda t)^{-\alpha}$$

MGF of $\underbrace{\sum_{n=1}^N X_n}_{\rightarrow Y}$ is given by

$$M_Y(t) = \prod_{n=1}^N (1 - \lambda t)^{-\alpha} = (1 - \lambda t)^{-N\alpha}$$

which is MGF of $\text{Gamma}(N\alpha, \lambda)$
 So, $Y = \sum_{n=1}^N X_n \sim \text{Gamma}(N\alpha, \lambda)$

$$\text{So } E Y = N\alpha\lambda$$

$$\text{Kor } Y = N\alpha\lambda^2$$

$$E \bar{X}_N = \frac{E Y}{N} = \alpha\lambda$$

$$\begin{aligned} \text{Kor } \bar{X}_N &= E X_N^2 - (E \bar{X}_N)^2 \\ &= \frac{E Y^2}{N^2} - \alpha^2 \lambda^2 \\ &= \frac{N\alpha\lambda^2 + N^2\alpha^2\lambda^2}{N^2} - \alpha^2 \lambda^2 \\ &= \frac{N\alpha\lambda^2}{N^2} = \frac{\alpha\lambda^2}{N} \end{aligned}$$

So, $\bar{X}_N$ follows a Gamma distribution with mean $\alpha\lambda$ & variance $\frac{\alpha\lambda^2}{N}$

Let $\bar{X}_N \sim \text{Gamma}(A, B)$

Then $E \bar{X}_N = AB = \alpha \tau$

$$\text{Var}(\bar{X}_N) = AB^2 = \frac{\alpha \tau^2}{N^2}$$

So, $A = N^2 \alpha$, $B = \frac{\tau}{N^2}$

So, $\bar{X}_N \sim \text{Gamma}(N^2 \alpha, \frac{\tau}{N^2})$

we know that

$$b_x(x) = \int_{y \in S_y} b_{x,y}(x,y) dy$$

observe that

$$b_x(x) = \int_0^\infty \underbrace{\frac{1}{y} \exp\left(-y - \frac{x}{y}\right)}_{\hookrightarrow b_{x,y}(x,y)} dy$$

$$b_{x|y}(x|y) = \frac{b_{x,y}(x,y)}{b_y(y)}$$

$$\Rightarrow b_{x,y}(x,y) = b_y(y) b_{x|y}(x|y)$$

$$\begin{aligned} \Rightarrow b_{x,y}(x,y) &= \frac{1}{y} \exp\left(-y - \frac{x}{y}\right) \\ &= \underbrace{\frac{1}{y} \exp\left(-\frac{x}{y}\right)}_{\hookrightarrow b_{x|y}(x|y)} \underbrace{\exp(-y)}_{\hookrightarrow b_y(y)} \end{aligned}$$

So,

$$x|y \sim \exp(y) \quad , \quad y \sim \exp(1)$$

$$\begin{aligned}
 3) \quad b_x(x) &= 2 \left(\frac{1}{2} \int_0^\infty \frac{1}{y} e^{(-y - \frac{x}{y})} dy \right) \\
 &= \int_0^\infty \frac{1}{y} e^{(-y - \frac{x}{y})} dy \\
 \int_0^\infty b_x(x) dx &= \int_0^\infty \int_0^\infty \frac{1}{y} e^{(-x - \frac{x}{y})} dy dx \\
 &= \int_0^\infty \frac{1}{y} e^{-x} \int_0^\infty e^{-\frac{x}{y}} dx dy \\
 &= \int_0^\infty \frac{1}{y} e^{-x} \left(y (0 - (-1)) \right) dy \\
 &= \int_0^\infty e^{-x} dy = 1
 \end{aligned}$$

$$X|Y \sim F, Y \sim G$$

$$b_{x|y}(x|y) = \frac{b_{x,y}(x,y)}{b_y(y)}$$

$$\rightarrow f(x,y) = f(x|y) f(y)$$

$$\begin{aligned}
 \int_0^\infty \int_0^\infty \frac{1}{y} e^{-x - \frac{x}{y}} dy dx &= f(x,y) \\
 \int_0^\infty \frac{1}{y} e^{-y} dy &= f(y) \\
 \int_0^\infty e^{-x/y} dx &= f(x|y)
 \end{aligned}$$

$$6) \quad Y_1, \dots, Y_r \mid X \stackrel{\text{iid}}{\sim} \text{Poisson}(x)$$

$$b_{Y|X}(y, \dots, y_r | x) = \begin{cases} \frac{\exp(-x) x^y}{y!}, & y \in \{0, 1, \dots\} \\ 0, & \text{o.w.} \end{cases}$$

$$X \sim \text{Gamma}(a, b)$$

$$b_X(x) = \frac{x^{a-1} b^{-a} \exp(-\frac{x}{b})}{\Gamma(a)}, \quad x > 0$$

$$\begin{aligned} b_X(y_1, \dots, y_r) &= \int_0^\infty \frac{\exp(-x) x^y}{y!} dx \\ &= \frac{\Gamma(y+1)}{y!} = (y+1) \end{aligned}$$

$$b_{X|Y_1, \dots, Y_r}(x | y_1, \dots, y_r) = \frac{b_{Y|X}(y_1, \dots, y_r | x) b_X(x)}{b_Y(y_1, \dots, y_r)}$$

$$= \frac{\frac{x^{a-1} b^{-a} \exp(-\frac{x}{b})}{\Gamma(a)} \frac{\exp(-x) x^y}{y!}}{y+1}$$

$$f(x/y, y) =$$

$$\frac{x^{\alpha+y-1} b^{-\alpha} \exp(-x(\frac{b+1}{b}))}{\Gamma(\alpha) \Gamma(y+1)}$$